GENERAL ZOOLOGY COURSE (100Z)

(2018-2019)

Part V: Animal Taxonomy

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Animal Taxonomy

•is the branch of zoology concerned with the description, identification, nomenclature and classification of animals.

Modern System:

o Animal kingdom was divided into **phyla** (plural of phylum)*.
o Each phylum was divided into smaller groups called **class**.
o Each class was divided into an **order**.
o Each order was divided into **family*** (families).
o Each family was divided into a **genus** (plural-genera)
o Each genus was divided into a **species**. (scientific name)

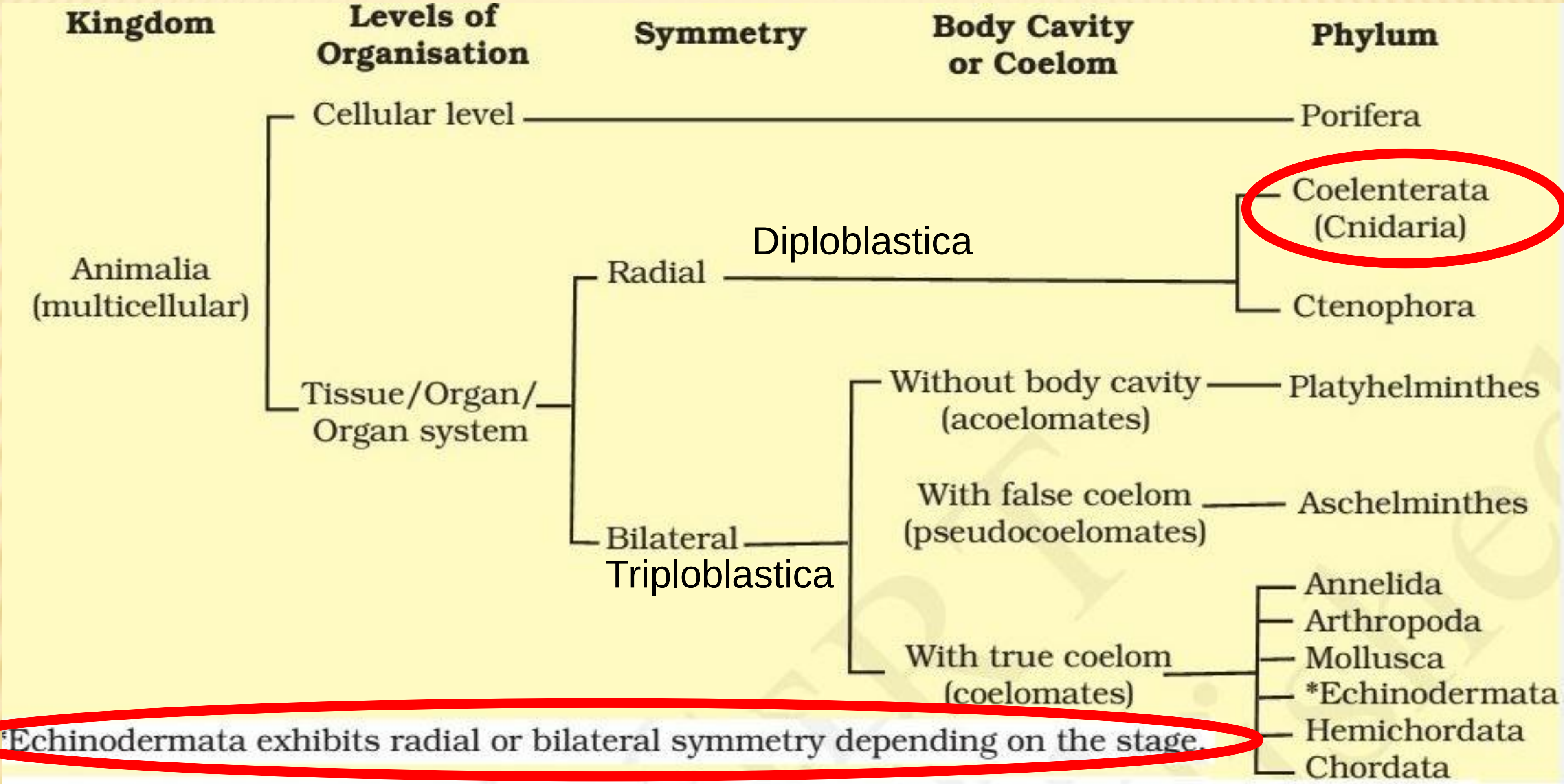
Kingdom
Phylum
Class
Order
Family
Genus
Species

√

The classification scheme

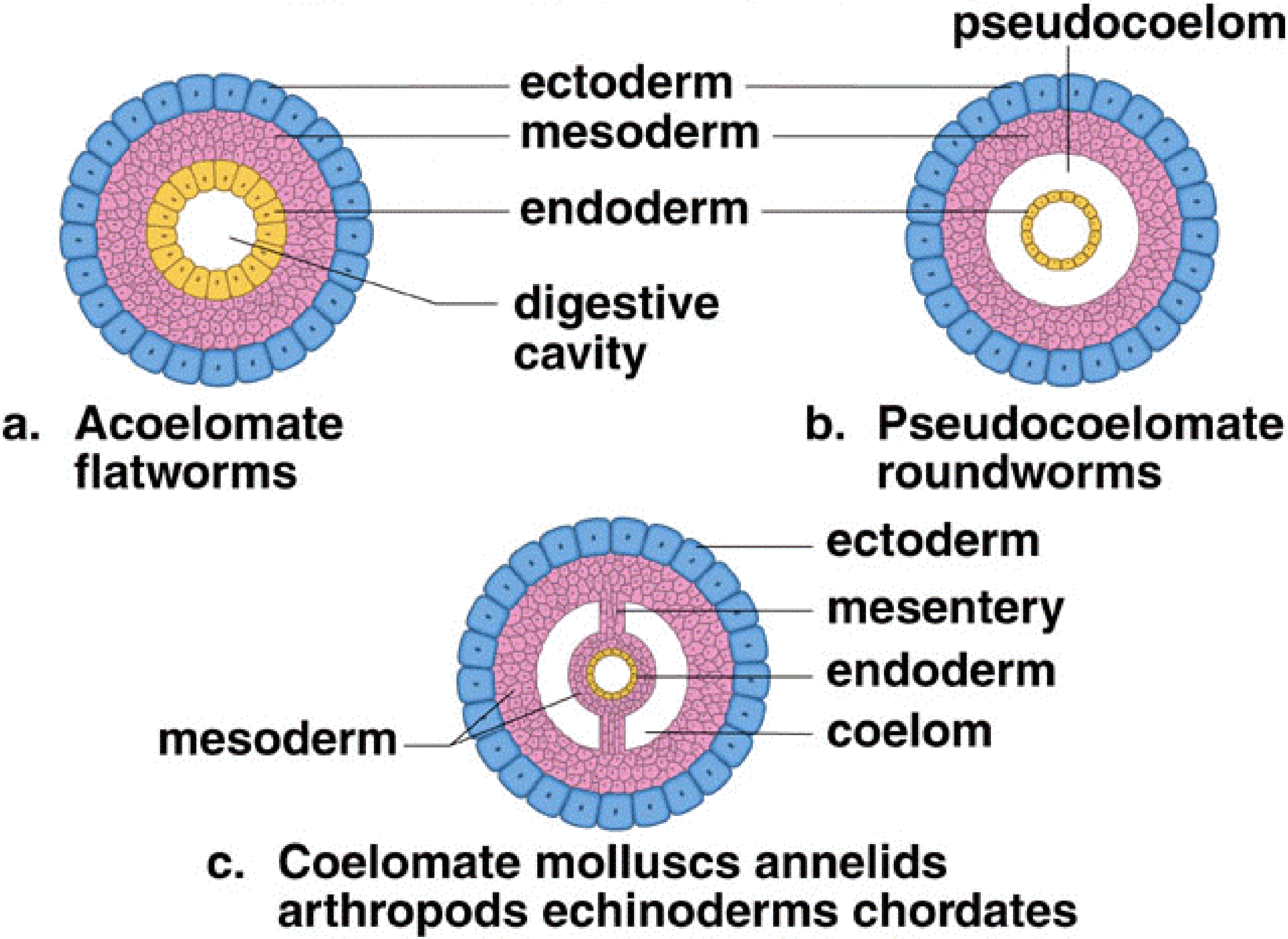
1- Kingdom Protista: Protozoan phyla (acellular organisms)
2- Kingdom Animalia (cellular animals)

- **Mesozoa**
  No tissues
- **Parazoa** نظائر البعديات
  no true tissues
  Porifera
- **Eumetazoa** البعديات الحقيقيه
  True tissues
  - **Diploblastica**
    Cnidaria
  - **Triplblastica**
    - **Acoelomata** → **Platyhelminthes**
    - **Pseudocoelomata** → **Nematoda**
    - **Coelomata** → **Annelida**, **Arthropoda**, **Mollusca**, **Echinodermata**, **Chordata**



*Echinodermata exhibits radial or bilateral symmetry depending on the stage.

Broad classification of Kingdom Animalia based on common fundamental features


pseudocoelom
ectoderm
mesoderm
endoderm
digestive cavity
a. Acoelomate flatworms
b. Pseudocoelomate roundworms
ectoderm
mesentery
endoderm
coelom
mesoderm
c. Coelomate molluscs annelids arthropods echinoderms chordates

What are general characteristics of protozoan phyla?:

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1-Unicellular;
some with multicellular stages in their life cycles.
2-No germ layer present.
3 -No organs or tissues,
but specialized organelles are found; nucleus single or multiple.
6-Locomotion by:
pseudopodia, flagella, cilia; some sessile.

7- Reproduction:-

a-) asexually by → الانشطار fission
→ التبرعم budding

b-) sexually → الاقتران conjugation
→ By gametes and meiosis

(union of male and female gametes to form a zygote).

Encystment التكيس:
During unfavorable environmental conditions (during dryness)
Encystment is a common feature of the life cycle of many protozoa where the animal secretes a resistant protective cyst around itself to enable it to survive unfavorable environmental conditions (during dryness) and for dispersal.
التكيس ظاهرة مميزة في الكثير من الأوليات تلجأ فيها الحيوانات الأولية إلى إفراز كيس واقيا حول أجسامها للحماية من الظروف الغير مناسبة.

Classification

Kingdom protista √

Subkingdom Protozoa

1- Phylum sarcomastigophora

1.1. **Subphylum Sarcodina:** moves by pseudopodia;

Class Lobosea Examples: *Amoeba* , *Entamoeba*.

1.2. **Subphylum: Mastigophora:** one or more flagella typically present in adult stages.

1.2.1. Class: Phytomastigophora, plant-like flagellates, usually bearing chromoplasts which contain chlorophyll. Examples: ***Euglena***.

1.2.2. Class: Zoomastigophora, flagellates without chromoplasts; one or more flagella. Examples: ***Trypanosoma***, ***Trichomonas***, ***Leishmania.***

2- Phylum Ciliophora: Cilia or ciliary organelles in at least one stage of life cycle; two types of nuclei. Examples: ***Paramecium***, ***Vorticella, Balantidium***

3-Phylum Apicomplexa:

- Characteristic set of organelles **(apical complex)** associated with anterior end present in some developmental stages;
- Class Sporozoa. Spores or oocysts typically present that contain infective sporozoites; one or two host life cycles. Examples: ***Plasmodium***, *Toxoplasma.*

Phylum porifera

General characteristics of porifera (sponges):

الاسفنجيات

1. Multicellular; body consists of a loose aggregation of cells.

2. Body with pores (ostia), canals, and chambers that serve for passage of water.

3. Mostly marine; all aquatic.

4. Excretion and respiration by diffusion.

5. Radial symmetry . التماثل شعاعى

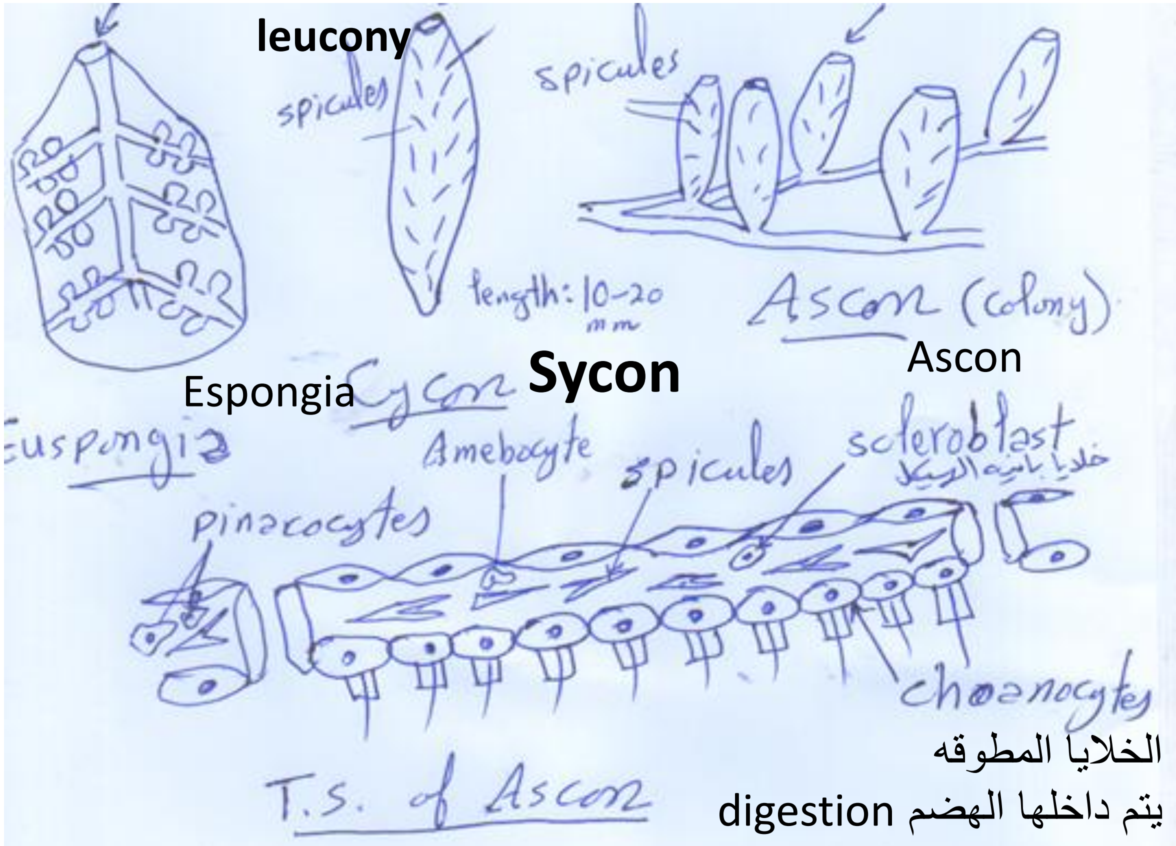
6. Skeletal structures include fibrillar collagen (a protein) and calcareous spicules.

7. There are no organs or no true tissues: and digestion is intracellular.

8. All adults are sessile and attached to substratum.

9. Asexual reproduction by buds and sexual reproduction by eggs and sperm; free swimming ciliated larva.

10. There are three patterns of sponges: Leuconoid (complex), Asconoid (colonies and single layer) and syconoid (double layer) patterns.



leucony

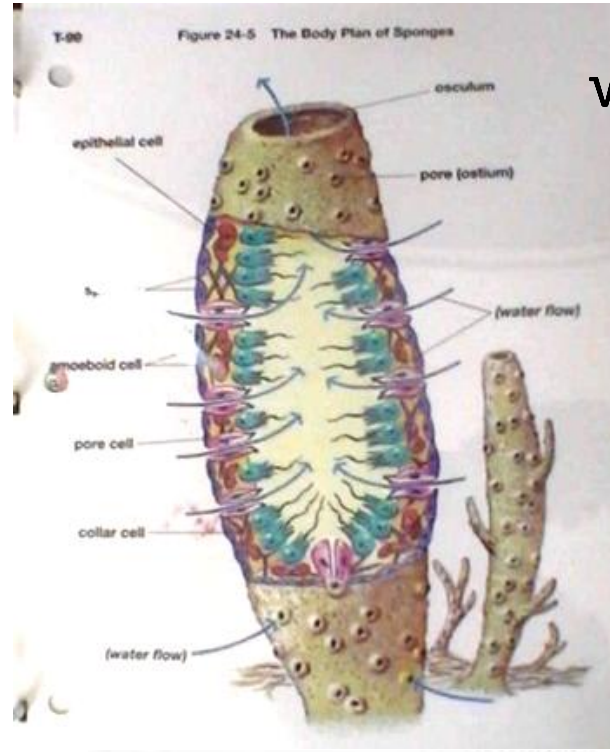
Espongia

Sycon

Ascon

الخلايا المطوقه

digestion يتم داخلها الهضم



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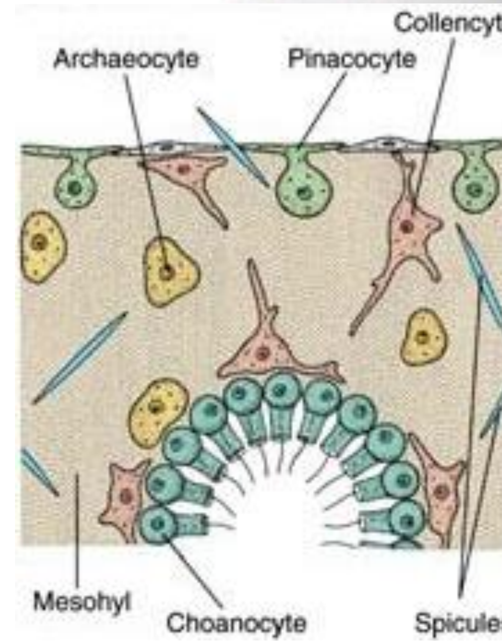
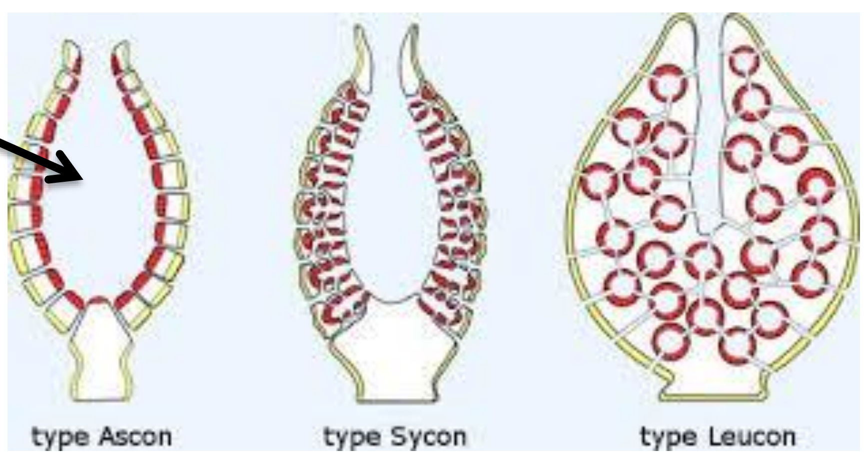


Spongocoel

تجويف نظير معوى



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1-Collagenous fibers
Found in euspongia

2- Calcerious spicules found in Ascon and Sycon

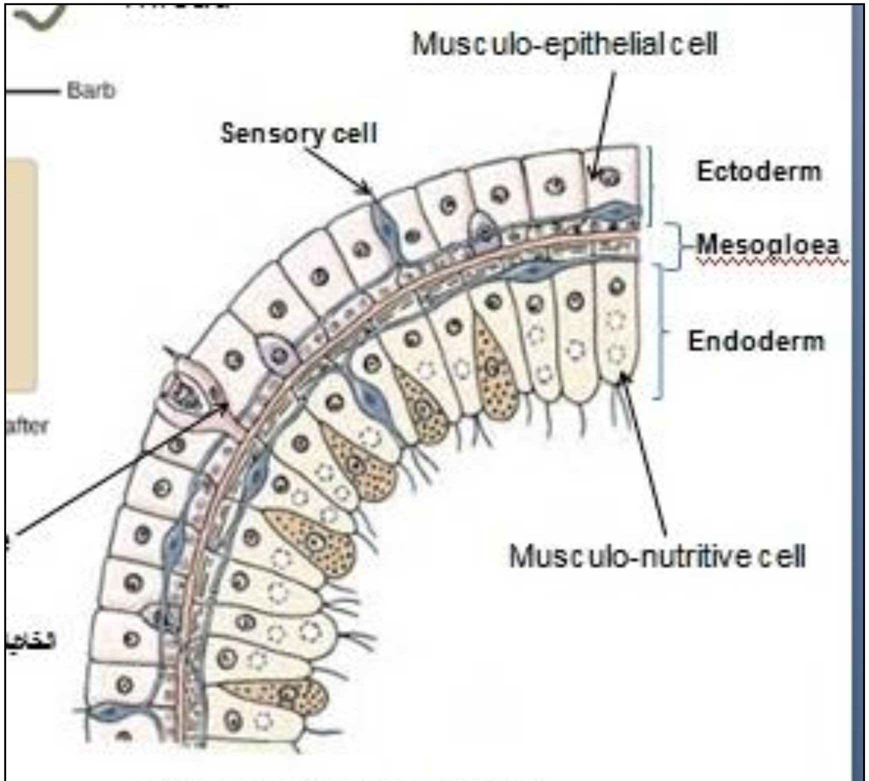
<u>Skeleton of sponges</u>

Nutrition:	**Organic matters found in water and digestion is interacellular.**

Phylum Cnidaria

General characteristics of Phylum Cnidaria : اللاسعات √

1) **Body with two layers, (diploblastic);** two well defined germ layers, ectoderm and endoderm; the third, mesogloea layer.



T.v. of Body wall of hydra

8) **Radial symmetry**

9) **Entirely aquatic**, some in fresh water but mostly marine.

4) **Two basic types of individuals:**

- **polyps (sessile, tubular in shape,** with the mouth facing the water upwards**)**
- **medusa (mobile, umbrella in shape,** with the mouth facing the water downwards**)** (jellyfish قنديل البحر).

5) **Polyp** produce either polyp or medusa by budding, while **medusa** produce only medusa

√

6) **Exoskeleton or endoskeleton:**

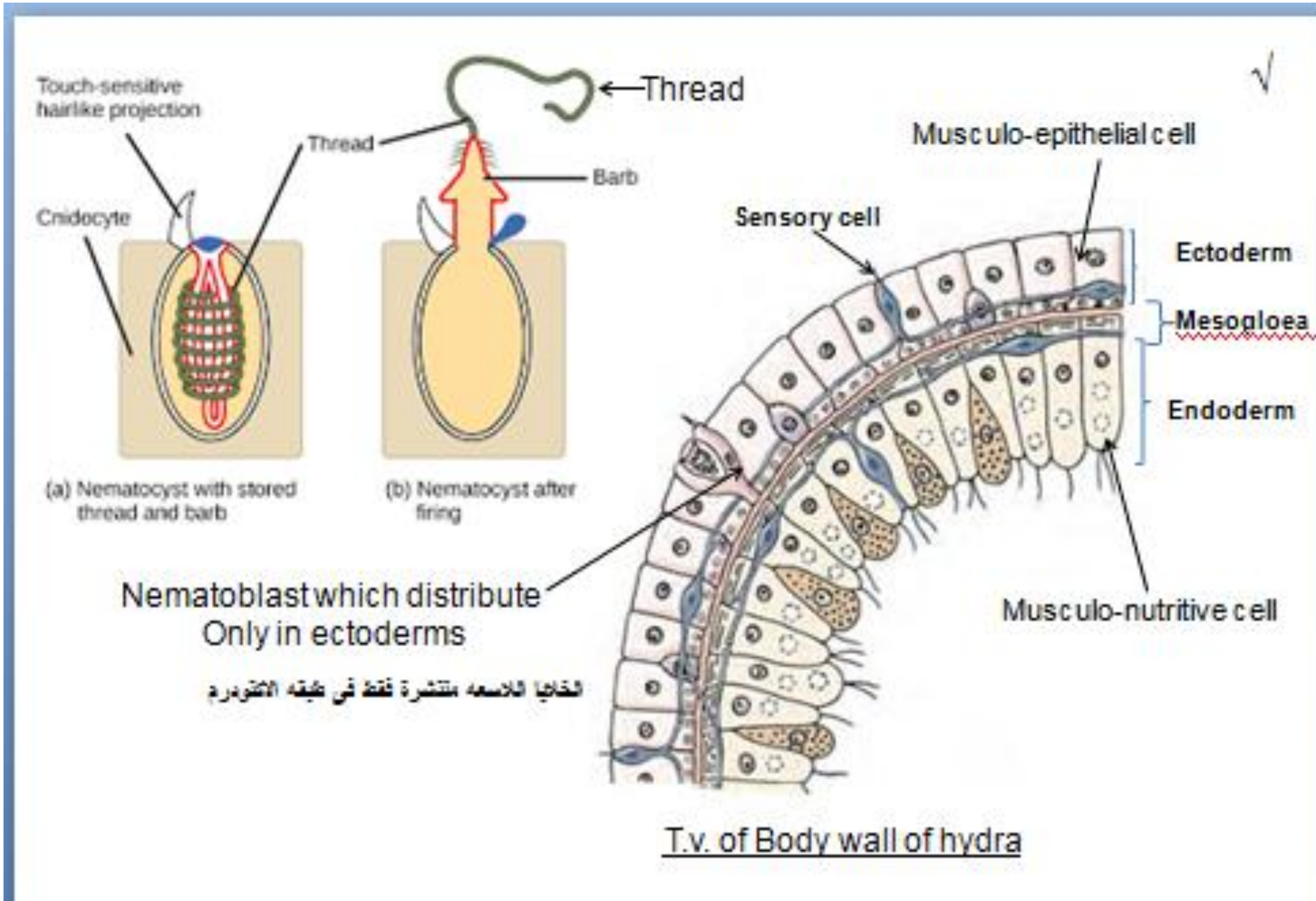
- of chitinous, calcareous components (Corals المراجين و الشعاب المرجانيه), or
- protein components in some.

7) **Gastro vascular cavity (Coelnteron)** تجويف جوفمعوى**:** (often branched or divided with septa) with a single opening that serves as both mouth and anus .

8) **There are** extensible tentacles usually encircling the mouth or oral region.

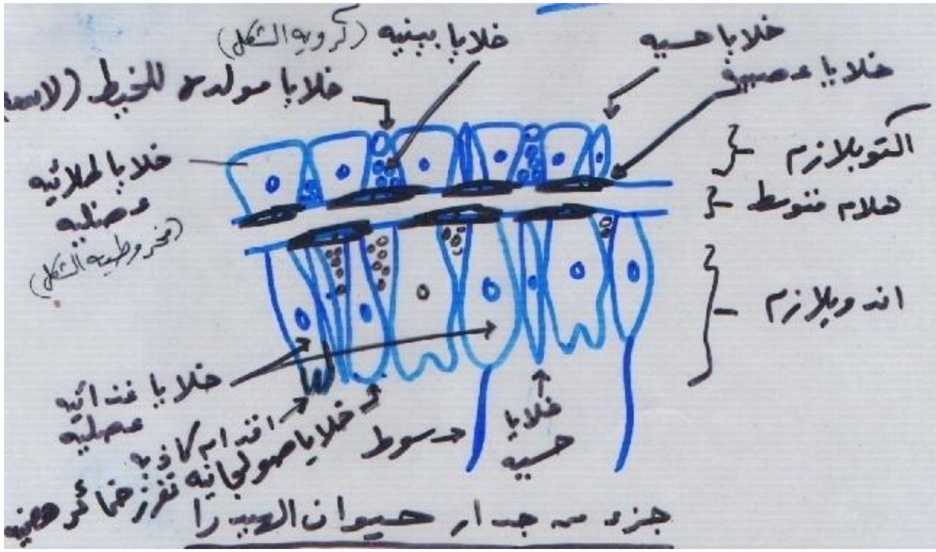
General characteristics of Phylum Cnidaria : اللاسعات

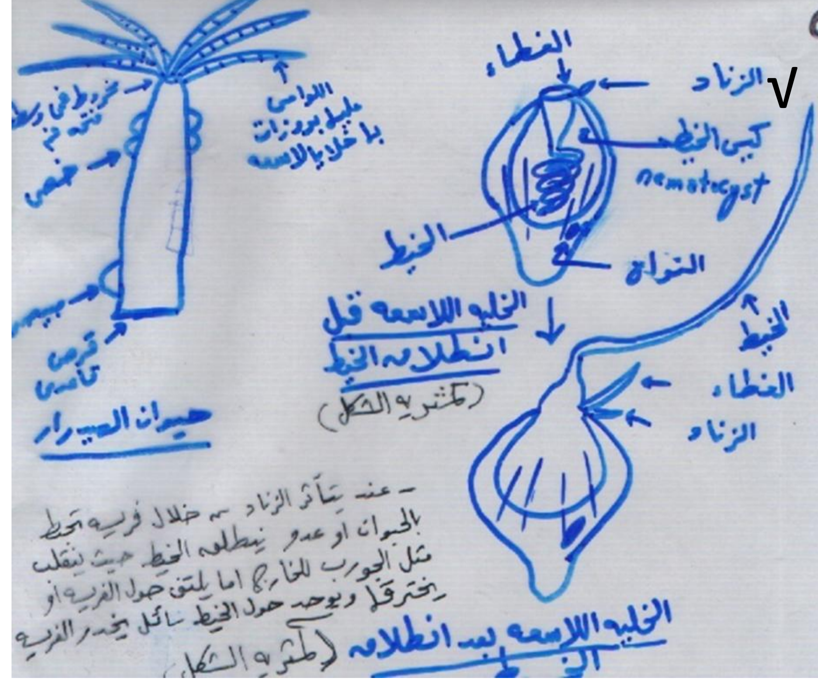
7) **there are nematocysts خلايا لاسعه** in either epidermis or gastrodermis or in both; nematocysts abundant **on tentacles,.**

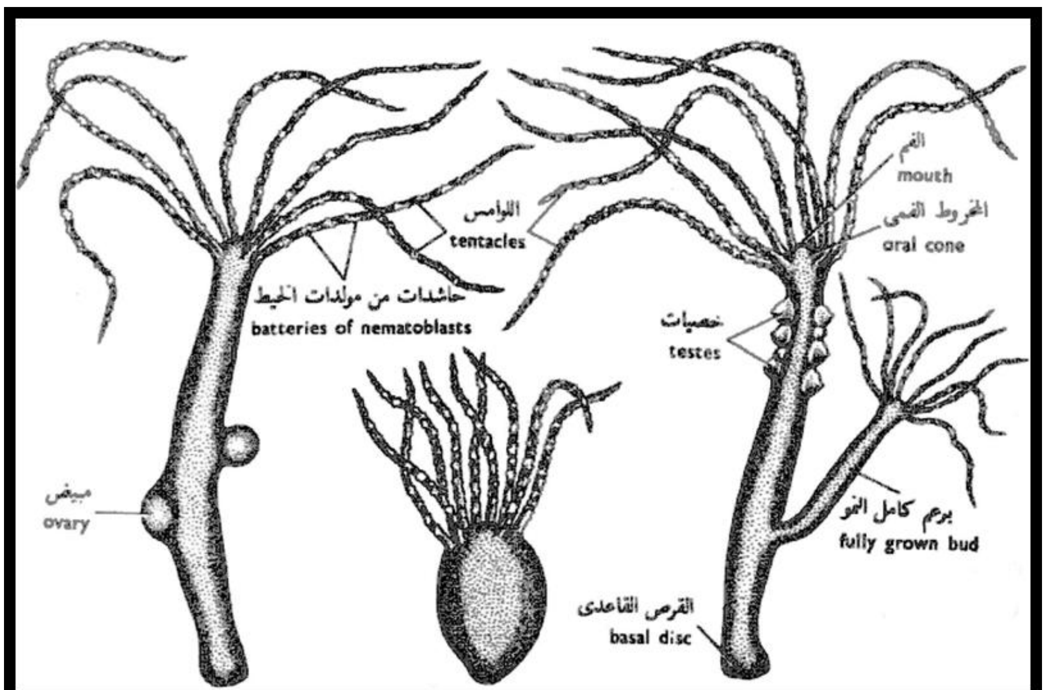


General characteristics of Phylum Cnidaria : اللاسعات

6) **Nerve net is found**, with symmetrical and asymmetrical synapses; with some sensory organs; diffuse conduction.
7) **Asexual reproduction by budding (in polyps) or sexual reproduction by gametes** (in all medusae and some polyps); sexual forms monoecious or dioeciously; **planula larva**; holoblastic indeterminate cleavage.
8) **No excretory or respiratory system (by diffusion).**


خلايا بينية (كروية الشكل)
خلايا حسية
خلايا عصبية
خلايا مولدة للخيط
اكتوبلازم
هلام متوسط
اندوبلازم
خلايا غذائية عضلية
خلايا حسية
جزء من جدار حيوان الهيدرا


الغطاء
الزناد
كيس الخيط
nematocyst
الخيط
النواة
الخليه اللاسعه قبل انطلاق الخيط
(كمثرية الشكل)
حيوان الهيدرا
الخليه اللاسعه بعد انطلاق الخيط


اللوامس
tentacles
حاشدات من مولدات الخيط
batteries of nematoblasts
مبيض
ovary
الفم
mouth
المخروط الفمي
oral cone
خصيات
testes
برعم كامل النمو
fully grown bud
القرص القاعدي
basal disc

Hydra

Class: Hydrozoa: *Hydra*

√

Hydra	
Living	Hydra lives in tidal rocks and aquatic plants in ponds البرك والمستنقعات in cool, clean fresh water of pools and streams. and wetlands.
Respiration:	Diffusion
Excretion:	Diffusion
Nutrition:	*Hydra* is a holozoic animal. It feeds on a variety of small crustacean, insect larvae and similar small aquatic animals.
Movement	Via swimming , gliding on its basal disc bu sereory mucus and floating

Reproduction

The *Hydra* reproduces sexually and asexually. Beside, it has a great capacity for regeneration.

Reproduction of hydra:

- **Asexual R.**
  - **Budding**
  - **Regeneration**
- **Sexual R. by formation gametes.**

Phylum Platyhelminthes

General characteristics of platyhelminthes:

المفلطحات

√

1. **Three germ layers (triploblastic). ثلاثيه الطبقات**

2. **Bilateral symmetry**; definite polarity of anterior and posterior ends. تماثل جانبى

3. **Body flattened** dorsoventrally; oral and genital apertures mostly on ventral surface. الجسم مفلطح

6. **acoelomate: لاسيلوميه** No internal body space other than digestive tube (acoelomate); spaces between organs filled with parenchyma, a form of connective tissue or mesenchyme.

7. **Digestive system incomplete (gastrovascular type); absent in some (in cestoda الديدان الشريطيه لايوجد قناه هضميه). لايوجد است**

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8. **Nervous system is found:** consisting of a pair of anterior ganglia with longitudinal nerve cords connected by transverse nerves and located in the mesenchyme in most forms; similar to cnidarians in primitive forms.

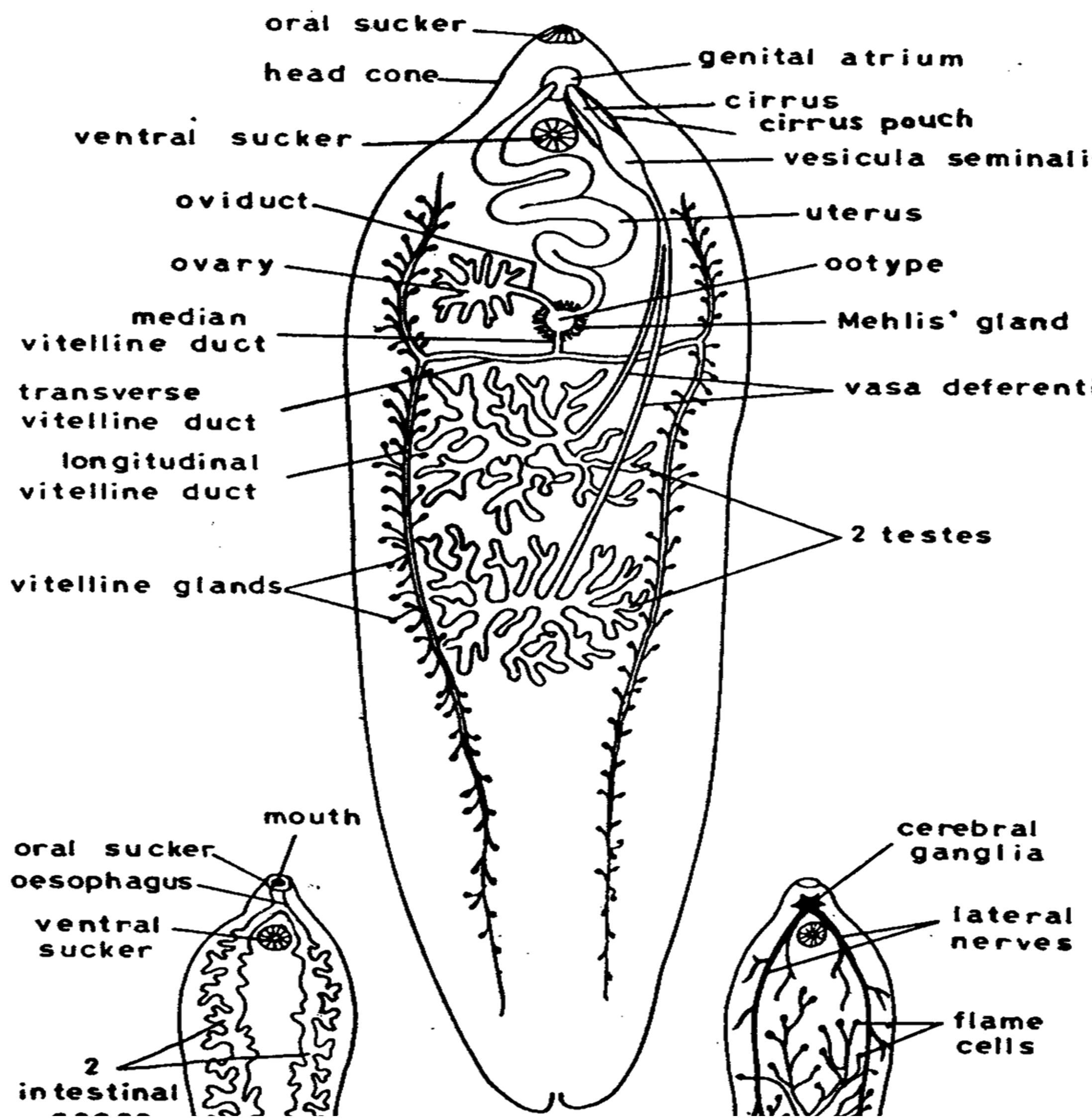
Intestinal Flukes

Fasciola is a widespread hepatic worm in Egypt <u>that lives in the bile ducts of the herbivores, especially cattle and sheep, and rarely affects humans</u>. It is a leaf-shaped flat worm that is between 2 and 8 cm long and 3-12mm width.

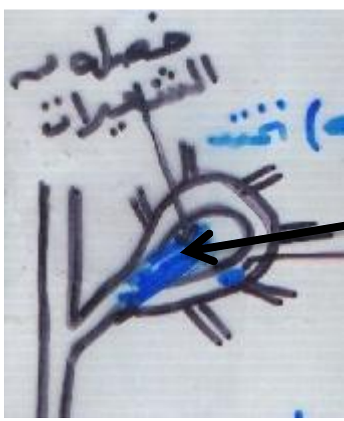
<u>***Fasciola*** feeds on its juices , blood, and tissues of the host.</u>

- ❑ ***Fasciola gigantica***
- ❑ **It is the common liver flukes in Egypt. It lives in the bile ducts of herbivorous animals. It is distributed in Egypt and some parts in Africa, Middle and Far-East.**
- ❑ **Large trematode measuring 2-7 cms, in length and 3-12mm in width. It is composed of an anterior small conical portion and a posterior larger leaf shaped portion with parallel sides. The cuticle is provided with numerous small spines.**
- ❑ **The ventral sucker is situated at the junction of the anterior with the posterior portions of the body and is slightly larger than the oral sucker which is close by at the extreme anterior end.**

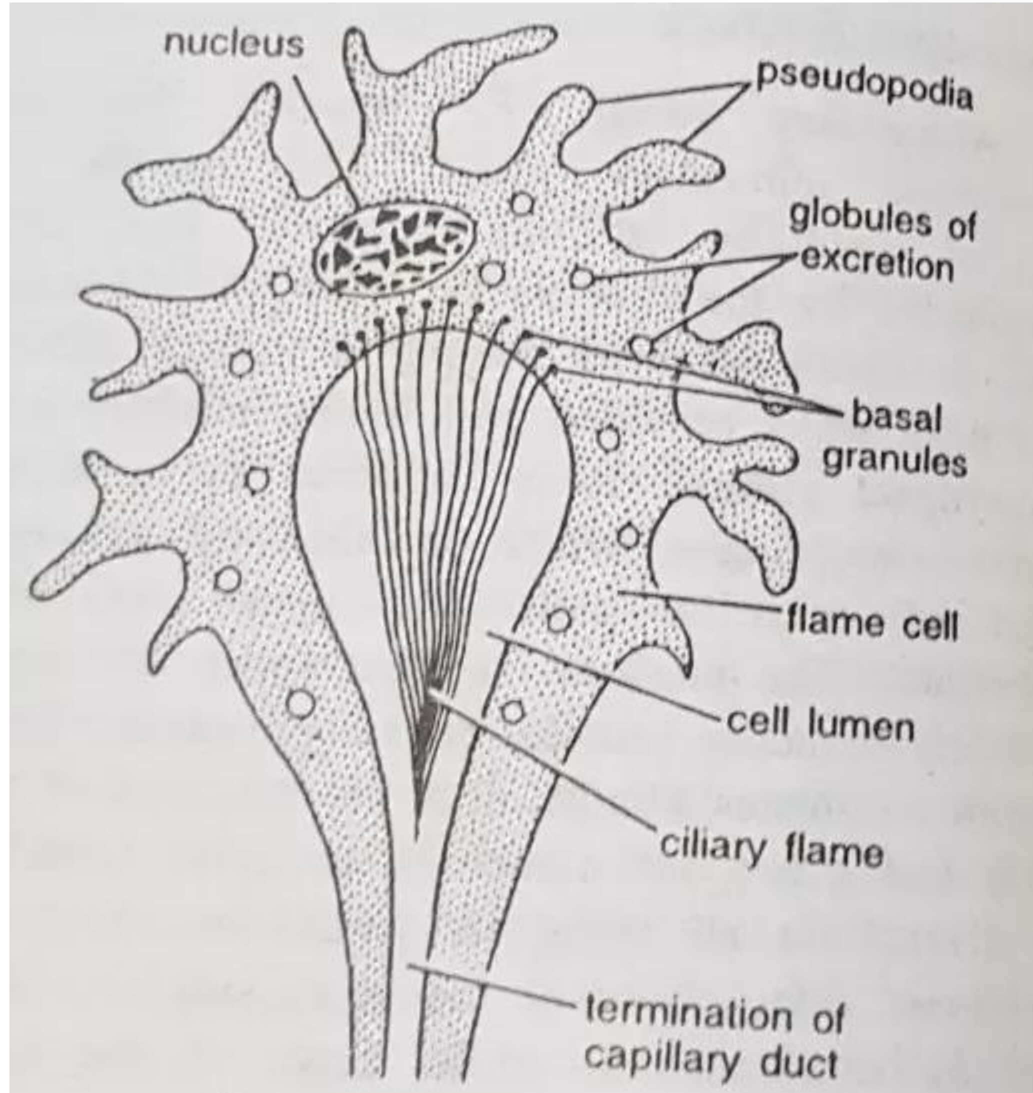
❑ ***Fasciola gigantica*** الدوده الكبديه



Length- 2-8 cm, width= 3-12 mm



Tuft of cilia



Flame Cell

	Fasciola gigantica	
	Shape: Large trematode measuring 2-7 cms, in length and 3-12mm in width.	
1	Living	**It lives in the biliary tract of sheep, cattle, rabbit, buffaloes, dog, and rarely man (final host).intermediate host: Lymnaea cailliaudi ….. Causin decaying liver disease.**
2	**Respiration:**	Via body surface or diffusion.
3	**Excretion:**	Flame cells which finally leads to main excretory canal.
4	**Nutrition:**	Adult: feeds on host tissues and juices. Cercaria: on decaying substances of plants in water.
5	**Movement**	
6	**Infective stage**	**Found attached to leaf plants. The cercariae are now infective and called the encysted metacercariae.**
7	**Mode of infection**	Circaria after escaping from the snails, adhere the aquatic plants and then lose their tail and secrete the glandular glands, to form encyst **Metacercaria**, which is infective stage, or this may occur on the water surface., By swallowing this infective stage which were attached To leaf plants like Watercress and lettuce الخس والجرجير

Phylum Nematoda

General Characteristics of Nematoda:

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1- They are elongated, cylindrical and triploblasitic.

2- Body wall with thick cuticle جليد, with no cilia.

3-The digestive system is complete with mouth, enteron, and anus.

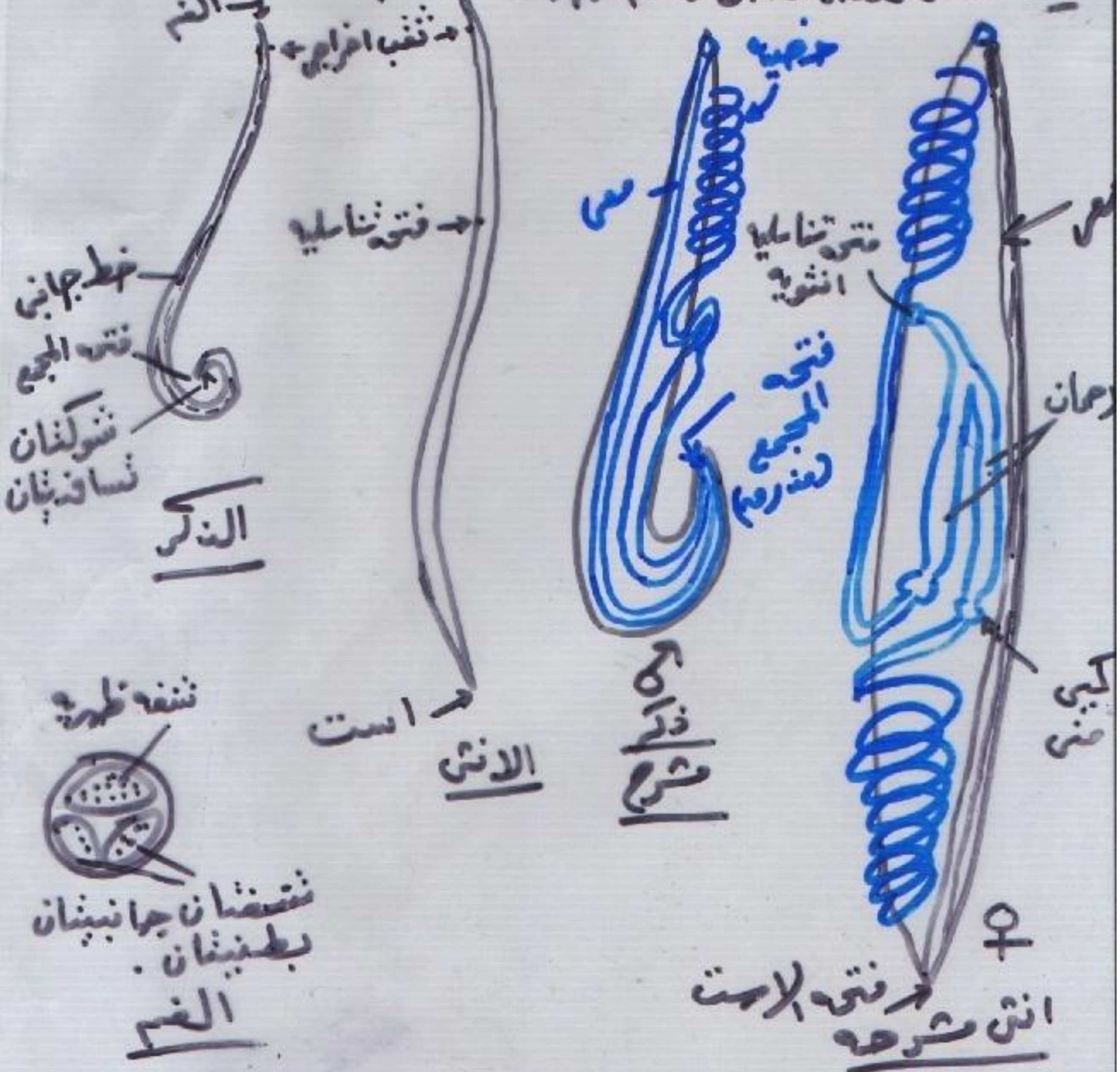
4- Sexes are usually separate الاجناس منفصله, with the female generally larger than the male.

9-Fertilization is internal, development is usually direct, but there may be a life history

	Ascaris lumbricoides	
	Shape: **Length: The length of the male between 15-25 cm and diameter of about 3 millimeters, while the female length of between 20-35 cm and diameter of 5 millimeters.**	
1	**Living**	**Ascaris lumbricoides_ are the most common nematode worms, which usually live in the small intestine of the human**
2	**Respiration:**	**There is no respiratory system . It is anaerobic respiration.** **(The majority of parasites do not use the oxygen available within the host, but employ systems other than oxidative phosphorylation for ATP synthesis.)**
3	**Excretion:**	**laterall canals system**
4	**Nutrition:**	**Adult: feed on digested food in the intestines.** **Larvae: feed on bacteria and organic matter and grow rapidly**
5	**Movement**	
6	**Infective stage**	**The egg containing larvae after the first molting becoming second-stage larvae. The eggs pass out with host's faeces.**
7	**Mode of infection**	**When the eggs containing the second stage larvae are swallowed by the host, the larvae hatch in the small intestine. By eating contaminated fresh vegetables without washing them thoroughly, or drinking water or dust polluted with this infective stage.**

الاسكارس

تتغذى على الغذاء المهضوم في الامعاء.



Phylum Annelida

General Characteristics of Annelida:

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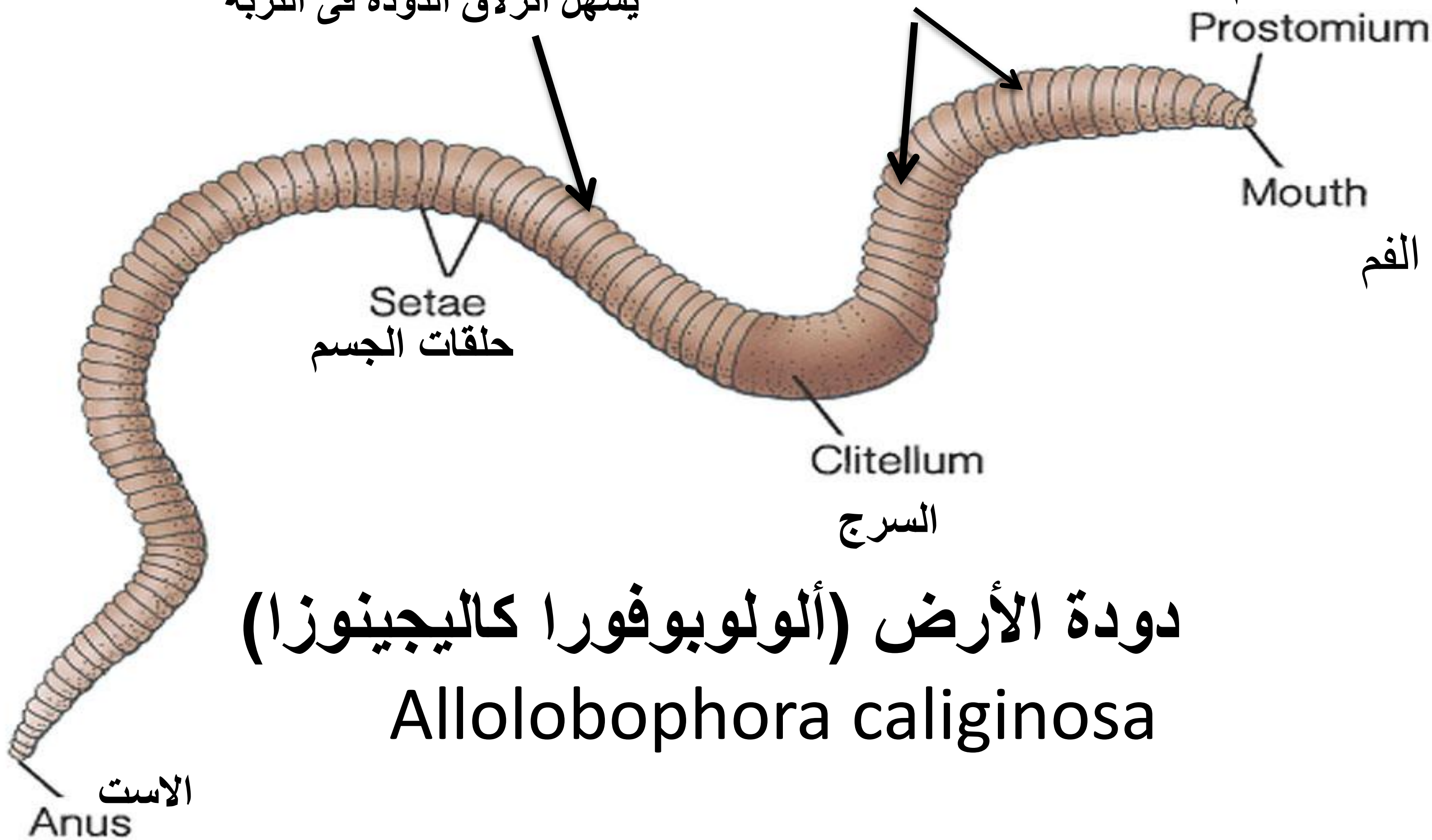
1. Triploblasic, Symmetry bilateral; coelomate. body long and conspicuously segmented both internal and externally.
2. Body divided into number of annuli حلقات (segmented body) and covered by moist cuticle over a sensory glandular epithelium.
3. Body cavity (coelom) well developed (except leeches) and usually subdivided by septa.
4. Digestive tract complete tubular, extending length of body.
5. Closed Circulatory system is found., longitudinal vessels, with branches to each segment; blood plasma with hemoglobin.
6. Excretory system typically a pair of nephridia per body segment.
7. Most are hermaphrodite خناث, external mixed fertilization, sexes united and development direct or indirect.



دودة الأرض (ألولوبوفورا كاليجينوزا)

Allolobophora caliginosa

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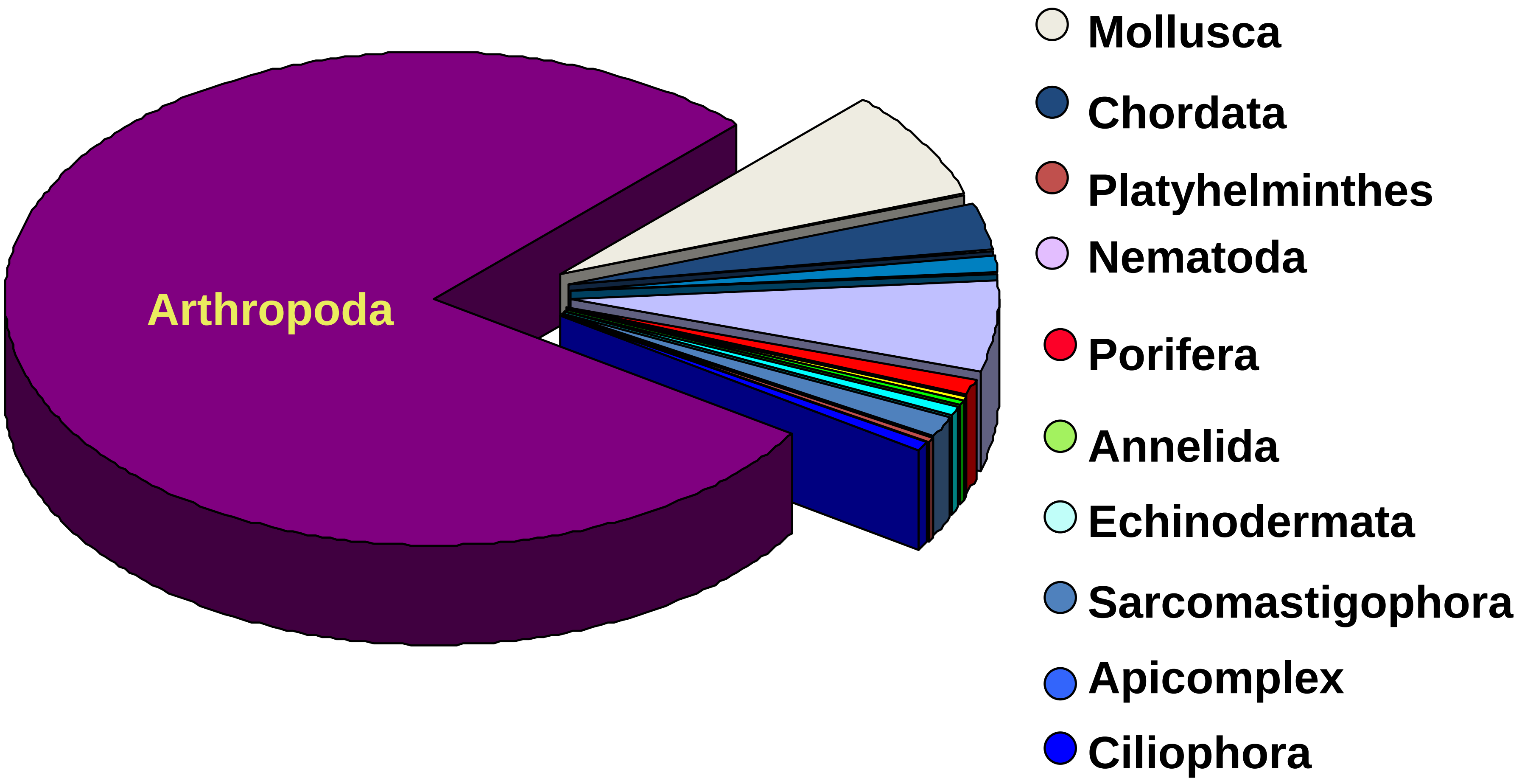
Phylum Arthropoda

Phylum Arthropoda (Gr. Arthron: joint + podos: foot).

Jointed Legs

Number of species



General Characteristics

1- Bilateral symmetry with metameric body divided into tagmata consisting of head and trunk; head, thorax and abdomen; or cephalothorax and abdomen.

2- With jointed appendages; - one pair to each somite

- number 10 reduced appendages often modified for specialized functions.

3- Exoskeleton consists of cuticle containing protein, lipid chitin, and often calcium carbonate secreted by underlying epidermis and shed (molted) at intervals.

Phylum: Chordata

Phylum Chordata

Characteristic:

1-The presence of a notochord:

The notochord is a long flexible, rod-like supporting structure found at all chordates. The name of the whole phylum is derived from this feature.

هو تركيب طويل مرن عصوى مدعم يوجد فى كل الحبليات

The notochord either persists throughout the whole life, or may only appear in the embryonic stages and then becomes gradually replaced by the vertebral column. The latter case is found in higher chordates, namely the vertebrates

وهو اما يوجد طول فترة عمر الحيوان أو يوجد فى المراحل الجنينية فقط ويستبدل بعد ذلك تدريجيا بالعمود الفقرى كما فى الفقاريات

2- A dorsal nerve cord:

- A tubular nerve cord which runs dorsal to the notochord (chordates).

- in the invertebrates the nerve cord extends along the ventral side of the body.

-the nerve cord in the invertebrates is solid .
- it is hollow in the chordates.
- The internal cavity of the nerve cord is known as the central canal.
- The anterior part of the nerve cord becomes enlarged to form the brain

3-Pharyngeal pouches and gill-slits:

-These pouches open externally by narrow opening known as the gill-slits.

- Such slits either remain throughout the whole life of the animal.

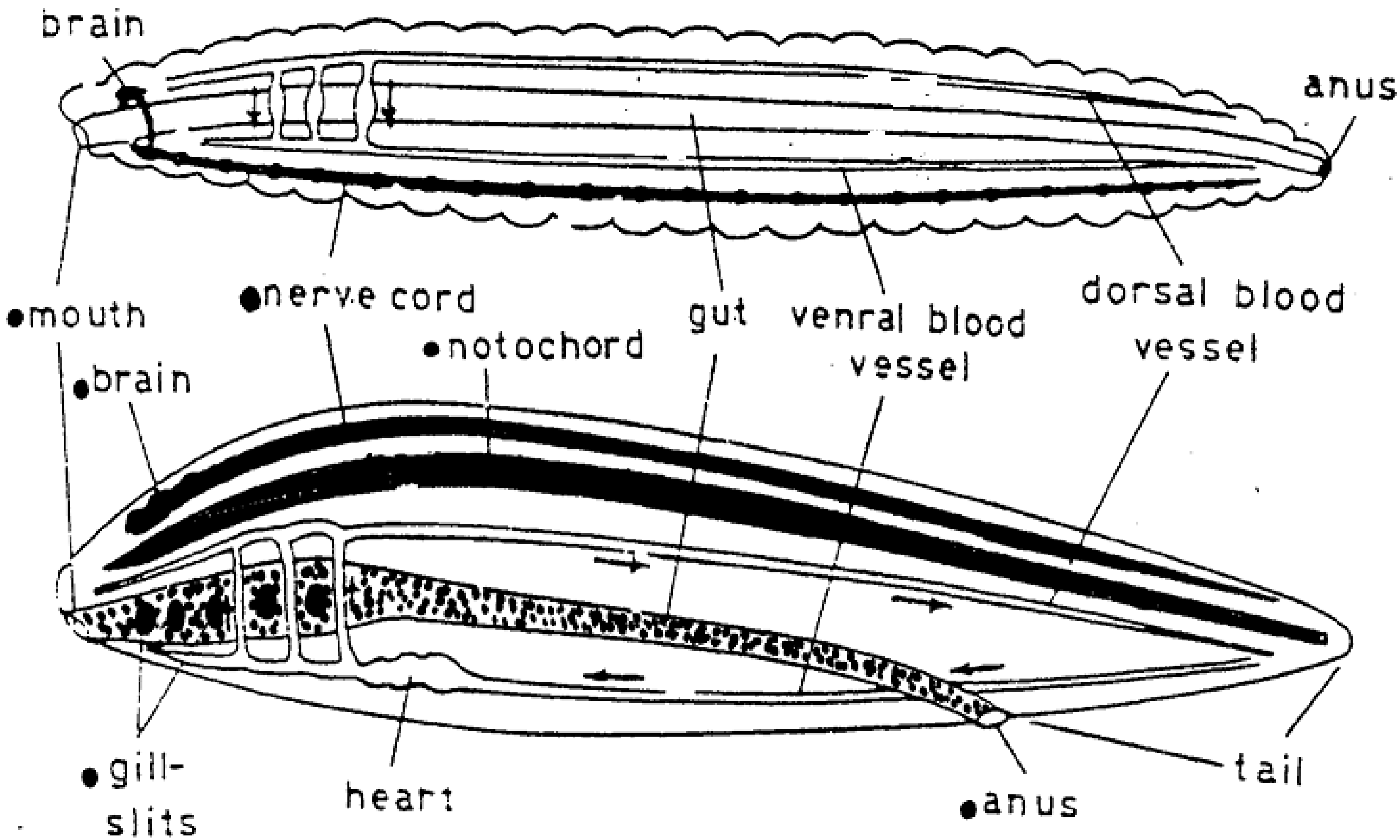
- Used for respiration as in fishes.

- may only appear in the early embryonic stages, then become closed gradually during embryonic development, and lungs develop instead for respiration

4-Direction of blood flow in the dorsal and ventral blood vessels:

- The blood flows backward in the dorsal blood vessel and forwards in the ventral blood vessel.
- This is contrary to what occurs in the invertebrates.

- 5- Presence of tail



Comparison between
an invertebrate (earthworm) and a chordate (fish)